Adaptive-QuantRL: Controlled Empirical Evaluation of Regime-Adaptive PPO/SAC Ensembles and Risk-Penalized Rewards in Dynamic Asset Allocation

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***Abstract*—Applying deep reinforcement learning (DRL) to financial portfolio management is frequently undermined by look-ahead leakage, unrealistic execution assumptions, uncalibrated risk penalties, and the absence of exposure-controlled baselines. In this paper, we present Adaptive-QuantRL, a strictly controlled 8-phase empirical framework that evaluates on-policy Proximal Policy Optimization (PPO) and off-policy Soft Actor-Critic (SAC) agents for dynamic allocation on unadjusted SPY price series from 2010 to 2024. We enforce a two-stage accounting mechanism (t close observation $\rightarrow t+1$ open execution with overnight gap reflection $\rightarrow t+1$ close valuation) under a one-way transaction cost of 0.1%, decouple log-normalized observation variance from raw non-negative reward variance, and evaluate 30 independently trained models across 5 random seeds (11, 22, 33, 44, 55). Out-of-sample testing (2023–2024) reveals three central findings: (1) SAC policies substantially suppress turnover, incurring only \$189–\$208 in cumulative transaction costs over 501 trading days and achieving a Sharpe ratio of 1.6317, CAGR of 12.16%, and Maximum Drawdown (MDD) of 5.61%–5.87%, outperforming the Moving Average (20/50) benchmark across all metrics and maintaining a Sharpe ratio above 1.16 even when transaction costs quintuple to 0.5%; (2) in contrast, PPO exhibits excessive turnover (\$1,078–\$1,131 in costs), which degrades both standalone PPO performance (Sharpe 0.33–0.54) and the validation-selected ADX(14) regime-adaptive ensemble (`ADX30_S75_W25`, Sharpe 1.4269), causing the ensemble to underperform a newly introduced *Exposure-Matched Baseline* (47.98% fixed equity weight, Sharpe 1.7170) by -0.2901 in Sharpe and -1.58%p in CAGR; and (3) calibrating the drawdown penalty coefficient ($\lambda_d = 0.02$, Reward B) reduces SAC's out-of-sample MDD in 4 out of 5 random seeds while cutting cross-seed Sharpe variance by 41%.**

***Index Terms*—Deep Reinforcement Learning, Portfolio Optimization, Soft Actor-Critic (SAC), Proximal Policy Optimization (PPO), Transaction Cost Sensitivity, Reward Ablation, Exposure-Matched Baseline**

I. Introduction

Deep reinforcement learning (DRL) has attracted considerable interest as a framework for continuous portfolio allocation, promising to integrate nonlinear market signals with transaction-cost-aware execution [1]–[3]. Despite proliferating architectures, empirical claims in financial DRL often suffer from four methodological vulnerabilities:

1) **Information Leakage and Execution Mismatch**: Fitting feature scalers across the full sample or assuming instantaneous execution at the same closing price ($Close_t$) used to compute technical indicators introduces severe look-ahead bias.
2) **Confounded Beta Exposure**: Dynamic DRL agents typically hold an average equity weight between 40% and 60%. Comparing such agents solely against 100% Buy-and-Hold or noisy Moving Average crossover rules confounds genuine market-timing alpha with passive risk-reduction beta.
3) **Reward Scale Collapse and Variance Sign Inversion**: Adding uncalibrated drawdown or variance penalties to daily log returns often causes policy collapse into 100% cash. Furthermore, feeding z-score standardized variance into a reward penalty term reverses the penalty into a reward whenever variance falls below its historical mean.
4) **Single-Seed Fragility**: Reporting cherry-picked trajectories without multi-seed distribution metrics obscures initialization sensitivity and turnover fragility under higher friction.

To address these challenges, we design **Adaptive-QuantRL**, an institutional-grade experimental protocol that separates ensemble rule discovery from reward ablation across 30 independent checkpoints. Rather than selectively reporting favorable metrics, we rigorously test four pre-registered Research Questions (RQ1–RQ4) and transparently report both the strengths of off-policy entropy-regularized allocation (SAC) and the negative market-timing alpha of PPO-blended ensembles when benchmarked against an exposure-matched baseline.

II. Problem Formulation & Accounting Protocol

A. POMDP State Space and Strict Scaler Isolation

We formulate single-asset dynamic allocation between the S&P 500 ETF (SPY, unadjusted OHLC prices excluding dividends to isolate pure price-action behavior) and cash as a Partially Observable Markov Decision Process (POMDP). At each trading day t after market close, the agent observes a 483-dimensional state vector $s_t \in \mathbb{R}^{483}$ stacking a 30-day

lookback window of 16 market features ($30 \times 16 = 480$) and 3 current portfolio state variables:

$$s_t = \left[\tilde{X}^{\text{mkt}}_{t-29:t},\ w_t^{\text{close}},\ DD_t,\ \tilde{V}^{\text{obs}}_{30,t} \right]. \tag{1}$$

The 16 market features include multi-horizon returns ($\text{ret}_{1d}, \text{log_ret}_{1d}, \text{ret}_{5d}, \text{ret}_{20d}$), moving average divergences ($\text{Close}/\text{SMA}_{20,50,200} - 1$), 20-day realized volatility (vol_{20d}), RSI_{14}, normalized MACD components, normalized ATR_{14}, and directional movement indicators ($\text{ADX}_{14}/100, +\text{DI}_{14}, -\text{DI}_{14}$). All market features $\tilde{X}^{\text{mkt}}$ are standardized using mean and standard deviation statistics fitted *exclusively* on the Train split (2010–2019).

Crucially, we decouple the variance representation between the observation space and the reward function. Let $\sigma^{2,\text{raw}}_{30,t} = \text{Var}(r^{\text{port}}_{t-29:t}) \geq 0$ denote the 30-day rolling sample variance of daily portfolio returns (set to 0 for $t < 30$). The agent observes a log-normalized transformation:

$$\tilde{V}^{\text{obs}}_{30,t} = \frac{\ln(\sigma^{2,\text{raw}}_{30,t} + 10^{-8}) - \mu^{\ln V}_{\text{train}}}{\sigma^{\ln V}_{\text{train}}}, \tag{2}$$

whereas the reward function strictly accesses the non-negative raw variance $\sigma^{2,\text{raw}}_{30,t} \geq 0$, guaranteeing that risk penalties never turn positive during low-volatility regimes.

B. Two-Stage Execution and Accounting Identities

At day t close, the agent outputs a continuous target stock weight $a_t \in [0, 1]$. Execution occurs at day $t+1$ market open ($Open_{t+1}$), followed by end-of-day mark-to-market valuation at $Close_{t+1}$:

$$Equity^{\text{close}}_t = Cash_t + Shares_t \times Close_t, \tag{3}$$

$$Equity^{\text{open,pre}}_{t+1} = Cash_t + Shares_t \times Open_{t+1}, \tag{4}$$

where $Equity^{\text{open,pre}}_{t+1}$ incorporates the overnight gap return prior to rebalancing. To reach the target dollar allocation $S^{\text{target}}_{t+1} = a_t \cdot Equity^{\text{open,pre}}_{t+1}$ from current pre-trade holdings $S^{\text{pre}}_{t+1} = Shares_t \times Open_{t+1}$, the traded dollar notional is $\Delta S_{t+1} = |S^{\text{target}}_{t+1} - S^{\text{pre}}_{t+1}|$. With a one-way transaction cost rate $c = 0.0010$ (0.05% commission + 0.05% slippage), the deducted fee is $Cost_{t+1} = c \cdot \Delta S_{t+1}$, updating shares and cash as:

$$Shares_{t+1} = \frac{S^{\text{target}}_{t+1}}{Open_{t+1}}, \tag{5}$$

$$Cash_{t+1} = Equity^{\text{open,pre}}_{t+1} - S^{\text{target}}_{t+1} - Cost_{t+1}, \tag{6}$$

$$Equity^{\text{close}}_{t+1} = Cash_{t+1} + Shares_{t+1} \times Close_{t+1}. \tag{7}$$

Peak equity and end-of-day drawdown are tracked via $Peak_{t+1} = \max(Peak_t, Equity^{\text{close}}_{t+1})$ and $DD_{t+1} = (Peak_{t+1} - Equity^{\text{close}}_{t+1})/Peak_{t+1} \in [0, 1]$.

C. Calibrated Reward Ablation Formulations

We evaluate three hierarchical reward structures (R_A, R_B, R_C) at step $t \to t+1$:

$$R_{A,t} = \ln\left(\frac{Equity^{\text{close}}_{t+1}}{Equity^{\text{close}}_t}\right), \tag{8}$$

$$R_{B,t} = R_{A,t} - \lambda_d \cdot DD_{t+1}, \tag{9}$$

$$R_{C,t} = R_{B,t} - \lambda_v \cdot \sigma^{2,\text{raw}}_{30,t+1}. \tag{10}$$

To prevent excessive penalties from forcing all-cash policies, we measure pre-training reward component quantiles across three reference policies (Buy-and-Hold, Fixed 50:50, and Random Policy) over ten 252-day Train episodes. The pooled mean absolute log return is $\mathbb{E}[|R_A|] = 0.004285$ (median 0.002502), while pooled mean drawdown is 0.024126 and mean active 30-day raw variance is 4.857×10^{-5}. Selecting $\lambda_d = 0.02$ and $\lambda_v = 10.0$ calibrates the mean penalty terms to 11.26% and 11.33% of $\mathbb{E}[|R_A|]$, respectively.

III. Empirical Setup

A. Chronological Split and Controlled Training

Data spans January 2009 (warm-up buffer) through December 2024 for unadjusted SPY daily bars:

- **Train**: `2010-01-01` to `2019-12-31` (2,515 trading days; random 252-day episodes initialized at $10,000 cash).
- **Validation**: `2020-01-01` to `2022-12-31` (755 trading days; COVID-19 crash, 2021 bull market, and 2022 inflation bear market).
- **Test**: `2023-01-01` to `2024-12-31` (501 trading days; out-of-sample evaluation with strictly frozen models).

We train 30 independent models across two tracks: **Track 1 (Ensemble Track)** trains PPO and SAC with R_A across 5 random seeds $\{11, 22, 33, 44, 55\}$ (10 models); **Track 2 (Ablation Track)** trains PPO and SAC with R_B and R_C across the same 5 seeds (20 models). Every model is trained for exactly $10,080$ timesteps (40 episodes $\times$ 252 days) using identical '[128, 128]' MLP architectures, learning rate 3×10^{-4}, and $\gamma = 0.99$.

B. Shared-Env Dynamic Ensemble & Exposure-Matched Baseline

In Phase 5 and 6, we classify market regimes at day t close using $\text{ADX}_{14,t}$ against threshold candidates $\theta \in \{20.0, 25.0, 30.0\}$. If $\text{ADX}_{14,t} \geq \theta$, the regime is `Strong_Trend`; otherwise, `Weak_Trend`. For each seed $s \in \{11, 22, 33, 44, 55\}$, the action-level ensemble combines deterministic actions from frozen Reward A policies:

$$a^{\text{ens}}_t(s) = w^{\text{ppo}}_{\text{reg}} \cdot a^{\text{ppo}}_t(s) + (1 - w^{\text{ppo}}_{\text{reg}}) \cdot a^{\text{sac}}_t(s), \tag{11}$$

where $(w^{\text{ppo}}_{\text{strong}}, w^{\text{ppo}}_{\text{weak}}) \in \{0.25, 0.50, 0.75\}^2$, yielding $3 \times 3 \times 3 = 27$ candidate ensemble rules evaluated inside a shared Validation `TradingEnv`.

The top-ranked rule on Validation is frozen as `ADX30_S75_W25` ($\theta = 30.0$, $w^{\text{ppo}}_{\text{strong}} = 0.75$, $w^{\text{ppo}}_{\text{weak}} = 0.25$), which achieves a Validation mean Sharpe of 0.2267 ± 0.0846

and MDD of 18.98%. From this frozen ensemble's Validation trajectories, we extract each seed's average stock exposure: $\{11 : 48.26\%, 22 : 53.09\%, 33 : 44.60\%, 44 : 45.17\%, 55 : 48.76\%\}$ (mean 47.98%) and freeze these exact constant weights as the **Exposure-Matched Baseline** for out-of-sample testing.

IV. Empirical Results

A. RQ 1: Strategic Comparison Against Traditional Benchmarks

Table I and Table II report the Validation and Test performance across all evaluated models. During the 2023–2024 Test period (501 trading days), unadjusted SPY (`1_Buy_and_Hold`) rose 52.80% (CAGR 23.77%, volatility 12.88%, Sharpe 1.7212, MDD 10.29%). All three ADX threshold candidates ($\theta \in \{20, 25, 30\}$) exhibited statistically sufficient sample sizes ($N \geq 30$) in both strong-trend (332, 216, 128 days) and weak-trend (169, 285, 373 days) regimes.

Comparing the DRL algorithms reveals a stark structural divergence between SAC and PPO:

- **SAC Outperforms Active Rule-Based Trend Following**: `SAC_Reward_A` achieves a cumulative return of 25.63%, CAGR of 12.16%, Sharpe of 1.6317, and MDD of 5.87% at an average stock weight of 55.04%. Across all three reward variants, SAC decisively outperforms `4_Moving_Average_20_50` (CAGR 10.95%, Sharpe 1.0219, MDD 9.39%) by $+0.59$ to $+0.61$ in Sharpe ratio and reduces maximum drawdown by over 3.5%p. SAC also exceeds `3_Fixed_50_50` in cumulative return (25.63% vs. 24.05%) while maintaining nearly half the MDD of Buy-and-Hold (5.61%–5.87% vs. 10.29%).
- **PPO Suffers Severe Turnover Erosion**: Standalone PPO models (`PPO_Reward_A/B/C`) execute frequent day-to-day weight oscillations, accumulating $\$1,078.25$ to $\$1,131.39$ in transaction costs (roughly 11% of initial equity), compared to only \$189.00 to \$207.61 for SAC. Consequently, PPO's net CAGR collapses to 1.99%–4.03% (Sharpe 0.3286–0.5404).

B. RQ 2: Regime Behavior & Dynamic Ensemble vs. Exposure-Matched Baseline

In Phase 5 Validation analysis, PPO and SAC demonstrated low action correlation across all ADX regimes ($\rho = 0.0491$ overall; $\rho = 0.0128$ when $\text{ADX}_{14} \geq 30$), with PPO adopting a conservative mean allocation (35.34%) and SAC maintaining a balanced allocation (54.66%). In Phase 6, blending these complementary behaviors via `ADX30_S75_W25` reduced Validation MDD from 21.63% (SAC) and 20.99% (PPO) down to 18.98% and halved cross-seed Sharpe standard deviation to 0.0846.

However, out-of-sample Test evaluation (Table II) demonstrates why an **Exposure-Matched Baseline** is indispensable. While `Dynamic_Ensemble_ADX30_S75_W25` achieves a respectable Sharpe of 1.4269 ± 0.0755 and MDD of 5.81% (outperforming Moving Average 20/50 by $+0.4050$ in Sharpe), it trails the `Exposure_Matched_Baseline` (Sharpe 1.7170, CAGR 10.97%, MDD 5.04%) by -0.2901 in Sharpe and -1.58%p in CAGR. Because the ensemble assigns a 0.75 weight to PPO during strong trends ($\text{ADX}_{14} \geq 30$, 128 Test days), it inherits a portion of PPO's high turnover, paying \$466.21 in transaction fees versus \$13.44 for the passive exposure-matched portfolio. Thus, once average beta exposure (47.98%) is strictly controlled, the regime-switching ensemble does not generate positive net market-timing alpha in the 2023–2024 bull market.

C. RQ 3: Consistency of Risk Penalties Across Random Seeds (Reward Ablation)

Table III details the per-seed Test metrics for SAC across Rewards A, B, and C:

- **Drawdown Penalty ($R_A \rightarrow R_B$, $\lambda_d = 0.02$)**: In Validation (Table I), `SAC_Reward_B` ranked #1 overall, improving mean Sharpe ($0.2505 \rightarrow 0.2753$) and reducing mean MDD ($21.63\% \rightarrow 20.69\%$). In the Test split, `SAC_Reward_B` reduces mean MDD from 5.87% to 5.61% (-0.26%p) and lowers MDD in **4 out of 5 random seeds** (Seeds 22, 33, 44, and 55), reaching an MDD of 4.53% on Seed 33. Moreover, R_B reduces cross-seed Sharpe standard deviation by 41% ($0.1766 \rightarrow 0.1039$). For PPO, $R_A \rightarrow R_B$ lowers mean Test MDD by -0.20%p ($8.45\% \rightarrow 8.25\%$, improved in $2/5$ seeds).
- **Variance Penalty ($R_B \rightarrow R_C$, $\lambda_v = 10.0$)**: Adding the raw variance penalty in PPO slightly reduces annualized volatility ($7.34\% \rightarrow 7.32\%$, -0.02%p, $2/5$ seeds) and lifts mean Sharpe from 0.3286 to 0.5404. In SAC, where baseline policy volatility is already tightly bounded (6.93%), adding $\lambda_v = 10.0$ yields virtually identical volatility (6.97%, $+0.04$%p, improved/maintained in $3/5$ seeds) and Sharpe (1.6314 vs. 1.6077), indicating that drawdown penalization alone (R_B) suffices to regularize SAC's downside risk.

D. RQ 4: Reproducibility & Transaction Cost Sensitivity

To test robustness under market friction without retraining, we evaluate all frozen policies at elevated one-way transaction cost rates of $c = 0.3\%$ and $c = 0.5\%$ (Table IV).

- **SAC Resilience Under High Friction**: Even when transaction costs increase fivefold to 0.5% per trade, `SAC_Reward_A` maintains a strongly positive CAGR of 8.45%, Sharpe ratio of 1.1681, and MDD of 6.54%, outperforming `4_Moving_Average_20_50` across every cost tier ($+0.6098$ Sharpe at 0.1%, $+0.4631$ at 0.3%, and $+0.3175$ at 0.5%). Similarly, `SAC_Reward_B` and `SAC_Reward_C` preserve Sharpe ratios of 1.0741 and 1.1315 at $c = 0.5\%$.
- **Critical Breakdown of High-Turnover Policies**: PPO's net return turns sharply negative at $c = 0.3\%$ (`PPO_Reward_A` CAGR -8.00%, Sharpe -1.1851, cost $\$2,956.81$) and suffers severe capital depletion at $c = 0.5\%$ (CAGR -17.31%, MDD 32.64%, cost $\$4,458.73$). The dynamic ensemble outperforms Moving

TABLE I
PHASE 4 VALIDATION (2020–2022, 755 DAYS) AGGREGATED MODEL RANKING ACROSS 5 RANDOM SEEDS (ONE-WAY COST = 0.1%)

Rank	Algorithm	Reward	λ_d	λ_v	Cum. Ret (%)	CAGR (%)	Ann. Vol (%)	Sharpe (Mean ± Std)	MDD (Mean ± Std %)	Avg Weight (%)	Mean Cost ($)
1	SAC	Reward B	0.02	0.0	9.54	3.03 ± 2.77	14.28	**0.2753** ± 0.1761	**20.69** ± 2.36	53.38	314.80
2	SAC	Reward C	0.02	10.0	8.95	2.86 ± 2.46	14.37	0.2647 ± 0.1577	21.32 ± 2.59	51.58	302.35
3	SAC	Reward A	0.00	0.0	8.86	2.81 ± 2.73	14.84	0.2505 ± 0.1652	21.63 ± 1.12	54.66	291.05
4	PPO	Reward B	0.02	0.0	−4.39	−1.97 ± 7.70	14.75	−0.0429 ± 0.5261	22.05 ± 8.52	39.65	1,841.47
5	PPO	Reward A	0.00	0.0	−6.87	−2.40 ± 2.57	12.85	−0.1223 ± 0.1985	20.99 ± 2.60	35.34	1,766.98
6	PPO	Reward C	0.02	10.0	−7.41	−2.71 ± 4.55	13.97	−0.1544 ± 0.3669	20.97 ± 4.13	36.95	1,812.14

TABLE II
PHASE 7 OUT-OF-SAMPLE TEST (2023–2024, 501 DAYS) PERFORMANCE SUMMARY ACROSS BENCHMARKS AND 5-SEED RL MODELS (COST = 0.1%)

Category	Model / Strategy	Cum. Ret (%)	CAGR (%)	Ann. Vol (%)	Sharpe Ratio	Std Sharpe	Mean MDD (%)	Stock Weight (%)	Mean Cost ($)
Traditional Benchmark	`1_Buy_and_Hold`	52.80	**23.77**	12.88	**1.7212**	0.0000	10.29	100.00	10.00
Traditional Benchmark	`2_Cash_100`	0.00	0.00	0.00	0.0000	0.0000	0.00	0.00	0.00
Traditional Benchmark	`3_Fixed_50_50`	24.05	11.45	6.43	1.7171	0.0000	5.25	50.00	13.73
Traditional Benchmark	`4_Moving_Average_20_50`	22.94	10.95	10.73	1.0219	0.0000	9.39	73.65	95.86
Exposure-Matched Baseline	`Exposure_Matched_Baseline`	22.99	10.97	6.17	1.7170	0.0003	**5.04**	47.98	13.44
Single RL Model	`PPO_Reward_A`	4.72	2.31	7.03	0.3340	0.3636	8.45	32.78	1,094.15
Reward Ablation Model	`PPO_Reward_B`	4.01	1.99	7.34	0.3286	0.2296	8.25	33.07	1,078.25
Reward Ablation Model	`PPO_Reward_C`	8.27	4.03	7.32	0.5404	0.4638	8.73	33.67	1,131.39
Single RL Model	`SAC_Reward_A`	**25.63**	**12.16**	7.26	**1.6317**	0.1766	5.87	55.04	189.00
Reward Ablation Model	`SAC_Reward_B`	24.11	11.48	6.93	1.6077	**0.1039**	**5.61**	52.36	207.61
Reward Ablation Model	`SAC_Reward_C`	24.59	11.69	6.97	1.6314	0.1522	5.79	52.75	195.49
Dynamic Ensemble	`Dynamic_Ensemble_ADX30_S75_W25`	19.54	9.39	6.43	1.4269	**0.0755**	5.81	45.27	466.21

TABLE III
PER-SEED OUT-OF-SAMPLE TEST MDD (%) AND SHARPE RATIO FOR SAC REWARD ABLATION (RQ3 VERIFICATION)

Seed	Test MDD (%)			Test Sharpe Ratio		
	SAC-A	SAC-B	SAC-C	SAC-A	SAC-B	SAC-C
11	**5.80**	5.95	6.06	1.6016	**1.6431**	1.5134
22	7.72	**6.56**	7.41	1.4177	1.4392	**1.4499**
33	4.57	4.53	**4.38**	**1.8908**	1.6686	1.8146
44	5.31	**5.13**	5.17	1.6975	1.7038	**1.7419**
55	5.95	**5.87**	5.92	1.5509	1.5837	**1.6372**
Mean	5.87	**5.61**	5.79	**1.6317**	1.6077	1.6314
Std	1.16	**0.78**	1.12	0.1766	**0.1039**	0.1522

Average (20/50) at $c = 0.1\%$ ($+0.4050$ Sharpe) but crosses below Moving Average at $c = 0.3\%$ (-0.1810 Sharpe), confirming that blending high-turnover on-policy agents introduces substantial transaction-cost fragility.

V. DISCUSSION

Our empirical findings offer two methodological insights for quantitative RL researchers. First, algorithm selection in continuous action spaces directly governs portfolio turnover. On-policy PPO updates rely on local clipped surrogate objectives over recent rollouts, which in noisy financial environments induces oscillatory target weights unless an explicit turnover penalty is imposed. Off-policy SAC, by contrast, learns smooth Q-value surfaces over a replay buffer with entropy regularization, naturally moderating daily rebalancing.

Second, our results highlight the danger of evaluating RL ensembles without an *Exposure-Matched Baseline*. Had we compared `Dynamic_Ensemble_ADX30_S75_W25` (Sharpe 1.4269, MDD 5.81%) solely against `4_Moving_Average_20_50` (Sharpe 1.0219, MDD 9.39%) or standalone PPO (Sharpe 0.3340), we would have erroneously concluded that ADX regime switching generates strong alpha. Benchmarking against the frozen 47.98% constant-weight baseline (Sharpe 1.7170) reveals that the ensemble's low MDD stems from its conservative average exposure, while its active regime switching incurs transaction costs that outweigh timing gains during a persistent bull market.

Limitations: (1) All experiments use unadjusted SPY prices excluding dividend cashflows; while applied identically across all benchmarks, total return indices would shift absolute CAGR levels upward. (2) The Test split (2023–2024) represents a strong low-volatility bull market (SPY Sharpe 1.7212), where any cash buffer creates beta drag relative to 100% equity. Evaluating frozen models across multi-asset universes and prolonged bear markets is an important direction for future work.

VI. CONCLUSION

We presented **Adaptive-QuantRL**, a leak-free, accounting-verified benchmark for deep reinforcement learning in dynamic asset allocation. Across 30 models and 5 random seeds, off-policy SAC demonstrated superior cost efficiency and risk-adjusted performance (Test Sharpe 1.63, MDD 5.61%), consistently outperforming rule-based moving average strategies even under 0.5% transaction costs. Calibrated drawdown penalization (Reward B) improved downside protection in $4/5$ seeds, while our exposure-matched baseline framework provided a rigorous diagnostic separating beta exposure from net market-timing alpha.

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TABLE IV
PHASE 7 TRANSACTION COST SENSITIVITY ANALYSIS ON TEST SPLIT (2023–2024) ACROSS ONE-WAY COST RATES $c \in \{0.1\%, 0.3\%, 0.5\%\}$

One-Way Cost	Model / Strategy	CAGR (%)	Sharpe	MDD (%)	Mean Cost ($)	Excess Sharpe vs. Fixed 50:50	Excess Sharpe vs. Exp-Matched	Excess Sharpe vs. MA(20/50)
0.1% (Base)	3_Fixed_50_50	11.45	1.7171	5.25	13.73	0.0000	+0.0001	+0.6952
	4_Moving_Average_20_50	10.95	1.0219	9.39	95.86	−0.6952	−0.6951	0.0000
	Exposure_Matched_Baseline	10.97	1.7170	5.04	13.44	−0.0001	0.0000	+0.6951
	PPO_Reward_A	2.31	0.3340	8.45	1,094.15	−1.3831	−1.3830	−0.6879
	SAC_Reward_A	**12.16**	**1.6317**	5.87	189.00	−0.0854	−0.0853	**+0.6098**
	Dynamic_Ensemble_ADX30_S75_W25	9.39	1.4269	5.81	466.21	−0.2902	−0.2901	+0.4050
0.3% (3×)	3_Fixed_50_50	11.30	1.6969	5.27	41.14	0.0000	+0.0006	+0.7600
	4_Moving_Average_20_50	9.94	0.9369	9.75	285.18	−0.7600	−0.7594	0.0000
	Exposure_Matched_Baseline	10.83	1.6963	5.06	40.28	−0.0006	0.0000	+0.7594
	PPO_Reward_A	−8.00	−1.1851	18.09	2,956.81	−2.8820	−2.8814	−2.1220
	SAC_Reward_A	**10.29**	**1.4000**	6.19	556.58	−0.2969	−0.2963	**+0.4631**
	Dynamic_Ensemble_ADX30_S75_W25	4.81	0.7559	6.65	1,339.03	−0.9410	−0.9404	−0.1810
0.5% (5×)	3_Fixed_50_50	11.16	1.6765	5.29	68.49	0.0000	+0.0011	+0.8259
	4_Moving_Average_20_50	8.95	0.8506	10.12	471.35	−0.8259	−0.8248	0.0000
	Exposure_Matched_Baseline	10.69	1.6754	5.08	67.06	−0.0011	0.0000	+0.8248
	PPO_Reward_A	−17.31	−2.6041	32.64	4,458.73	−4.2806	−4.2795	−3.4547
	SAC_Reward_A	**8.45**	**1.1681**	6.54	910.71	−0.5084	−0.5073	**+0.3175**
	Dynamic_Ensemble_ADX30_S75_W25	0.43	0.0852	8.38	2,138.26	−1.5913	−1.5902	−0.7654

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